

Gene Pulser® Electroprotocols

* We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.

Cell Type	Mammalian, suspension	Molecules Electroporated	DNA: 6.5 kB, linearized plamid.
Species Used	Mouse, FDC-PI, II-3-dependent cell line		

Before the Pulse

Cell Growth Medium	RPMI 1640, 10% Fetal Calf Serum (GIBCO/BRL, Sigma)	Growth Phase at Harvest	Not given
		Pre-pulse Incubation	Ice, 10 min.

Wash Solution HEPES buffered saline (see notes)

The Pulse

Instruments Used Gene Pulser® apparatus

Electroporation Temperature	0 °C		
Electroporation Medium*	HEPES buffered saline	Cuvette Gap	0.4 cm
Cell Density	1 x 10 ⁷ (7) cells / ml	Voltage	1.5 kV
Volume of Cells	0.5 ml	Field Strength	3.75 kV/cm
DNA Concentration	1 mg / ml		
DNA Resuspension Buffer	TE Buffer (10 mM Tris, 1 mM EDTA, pH 8.0)	Capacitor	25 µF
Volume of DNA	10 µl (10 µg) / pulse	Resistor	(Pulse Controller) Ω none. NOT
After the Pulse		Time Constant	0.8 msec
Outgrowth Medium	RPMI 1640, 10% Fetal Calf Serum		

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

Ref: *EMBO J.* **9** : 4367-4374 (1990) Molecular cloning and expression of the murine interlenkin -5 receptor.
HEPES Buffered Saline: 140 mM NaCl, 5 mM KCl, 0.75 mM Na₂HPO₄, 6 mM dextrose, 25 mM HEPES, pH 7.2

**It is NOT RECOMMENDED to use high voltage with out the Pulse Controller.

Outgrowth Temperature	37 °C
Length of Incubation	Not given
Selection Method or Assay Used	G418, 400 µg / ml
Electroporation Efficiency	10 to 20 transfectants / µg DNA
Per Cent Survival	60% at 10 min. after electroporation

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